INCH-POUND

MIL-DTL-48412B
12 September 2000
SUPERSEDING
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DETAIL SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M53 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M53 Cartridge Actuated Initiator Remanufactured Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-A-2550 Ammunition, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Team (Code 840M), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by sending a letter.

AMSC N/A FSC 1377

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STANDARDS

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DEPARTMENT OF DEFENSE

MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

10520621	Initiator, Cartridge Actuated, M53, Metal Parts Assembly

(Application for copies should be addressed to Commander, Naval Surface Warfare Center, Indian Head Division, 101 Strauss Avenue, Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

PUBLICATIONS

AMERICAN SOCIETY FOR QUALITY CONTROL

ASQC Z1.4	Sampling Procedures and Tables for Inspection by Attributes
ASQC A8402	Quality Management and Quality Assurance Vocabulary

(Application for copies should be addressed to the American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate

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materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured initiator metal parts assembly shall contain a new shear pin and new O-rings.

3.2 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 14,000 to 15,000 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.3.1).

3.3 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration. Remanufactured exterior components (see 6.3.2) shall have the existing protective finish stripped and the specified protective finish reapplied.

3.4 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using the procedures of 4.5.3.

3.5 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 20 ± 1 pounds is applied at the steady rate of 0.125 inch per minute.

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.6.1 Integrated plant products. When remanufacturing and loading of the assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.6.2 Threaded component parts. Class 2A or 2B thread gage shall be used on remanufactured component parts inspection versus class 3A or 3B thread gage required on new hardware for initiators.

3.7 Workmanship. Initiators shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Conformance inspection (see 4.2).

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4.1.2 Quality assurance terms and definitions. Reference shall be made to ASQC A8402 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77°F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with ASQC Z1.4.

4.3.1.2 Size of lot. The production lot size shall be as specified in the contract.

4.3.2 Examination. On reprocessed parts, one hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.3.2.1 Examination of reclaimed parts. Examination of reusable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected.

4.3.2.2 Filter, Drawing 10520414, covering a detail of Drawing 10520621.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Nylon patch not properly located or fused	Manual/Visual
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr	Visual
105	Foreign matter	Visual

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Minor
None defined

4.3.2.3 Chamber, Initiator, Drawing 10520407, covering a detail of Drawing 10520621.

Critical
None defined

Major		
101	Pitch diameter of large external thread, min	Gage
102	Pitch diameter of small external thread, min	Gage
103	Pitch diameter of internal thread, max	Gage
104	Thread damaged	Visual
105	Metal defective	Visual
106	Protective finish inadequate or defective	Visual
107	Surface finish inadequate or defective	Visual
108	Burr	Visual
109	Foreign matter	Visual

Minor		
201	Surface finish improper, except as otherwise classified	Visual

4.3.2.4 Cap, Drawing 10520473, covering a detail of Drawing 10520621.

Critical
None defined

Major		
101	Pitch diameter of all thread, max	Gage
102	Thread damaged	Visual
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter	Visual

Minor
None defined

4.3.2.5 Pin, Shear, Drawing 10520622, covering a detail of Drawing 10520621.

Critical
None defined

Major		
101	Diameter, max	Gage
102	Total length, min	Measure
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual

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105	Surface finish improper	Visual
106	Burr	Visual
107	Foreign matter	Visual

Minor

201	Total length, max	Gage
202	Profile of end	Gage

4.3.2.6 Pin, Firing, Drawing 10520474, covering a detail of Drawing 10520621.

Critical
None defined

Major

101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Surface finish of body and striker tip improper	Visual
104	Burr	Visual
105	Foreign matter	Visual

Minor
None defined

4.3.2.7 Assembly, Drawing 10520621, less all Drawings except 10520557, 10520474, and MS29513-9.

Critical

1	Firing pin protrusion	Gage
2	O-ring missing	Visual

Major

101	Burr	Visual
102	Foreign matter	Visual

Minor

201	O-ring not lubricated	Visual

4.3.2.8 Assembly, Drawing 10520621, less Drawing 10520557.

Critical

1	Any part missing, except as otherwise classified	Visual

Major
None defined

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Minor		
201	Protective cap or plug missing	Visual
202	Protective finish inadequate or defective	Visual

4.4 Inspection testing. The following tests shall be performed in accordance with 4.5.

4.4.1 Hydrostatic pressure. Each chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.2 shall be cause for rejection of the subassembly.

4.4.2 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.3 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.5.3. Failure of any unit in the inspection sample to comply with 3.4 shall be cause for rejection of the assembly.

4.4.4 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.5 shall be cause for rejection of the lot.

4.5 Test methods and procedures.

4.5.1 Hydrostatic pressure. Prior to application of protective finish the cap and chamber shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.2.

4.5.2 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.3 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.5.4 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.5. The test shall be performed in a fixture in conjunction with a tensile testing machine.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points packaging activity

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within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems. As the Aircrew Emergency Escape Sytems are designed only for military aircraft, there is no commercial application for the M53 Cartridge Actuated Initiator Assembly

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and revision letter of this specification
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2 and 2.3) c. The responsible inspection activity (see 4.1)
c. Any special instruction requirements (see 4.3)
d. Level of packing and marking (see 5.1)

6.3 Definition.

6.3.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3.2 Exterior components. Remanufactured exterior components are: cap, firing pin housing, and chamber.

6.4 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.5 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian
Air Force - 99

Preparing activity
Navy - OS
(Project 1377-0138)

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Review activity
Air Force - 70

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-48412B	**2. DOCUMENT DATE (YYYYMMDD)** 20000912

3. DOCUMENT TITLE **INITIATOR, CARTRIDGE ACTUATED, M53 REMANUFACTURED METAL PARTS ASSEMBLY**

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION, NAVAL SURFACE WARFARE CENTER	b. TELEPHONE *Include Area Code)* (1) Commercial (301) 744-4700 (2) AUTOVON 354-4700
c. ADDRESS *(Include Zip Code)* STANDARDIZATION TEAM (CODE 840M) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533, Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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